Photosynthesis
Photosynthesis is the process plants use to turn light energy into chemical energy.
It takes place in the chloroplasts, which contain the green pigment chlorophyll.
The light-dependent reactions happen in the thylakoid membranes and produce ATP and NADPH.
Water is split during these reactions, releasing oxygen as a by-product.
The Calvin cycle happens in the stroma and uses ATP and NADPH to fix carbon dioxide.
The overall equation: 6CO2 + 6H2O + light energy -> C6H12O6 + 6O2.
Glucose made in photosynthesis is used for respiration or stored as starch.